TECHNICAL CATALOGUE · MOTOR-CUM-HAND DRIVEN

# CIE/777 HM

Mains motor with hand-crank operation

## Sustained-voltage insulation testing

A universal, multi-purpose instrument for checking insulation and conductance under simulated operating conditions in high-tension electrical and electronic fields. It is ideally suited to the electrical maintenance of rotating plant machinery and distribution networks, communication cables and insulators, and to the servicing of control and switching equipment, transformers and industrial installations. The dielectric absorption test needs the test voltage applied for several minutes, and operating the instrument under motor power makes this easy. Polarisation index is the resistance after 10 minutes divided by the resistance after 1 minute, and better results are obtained with longer test periods. The high sensitivity of the instrument gives sensible readings in these tests. The CIE/777 can save several hours in locating intermittent shorts, defective electrical connections, and insulation and conductor failure, and in preventing costly repairs and breakdowns. The CIE/777 HM can also be operated by hand cranking, with all other details as for the CIE/777.




CIE/777 HM · ILLUSTRATIVE IMAGE

## 01 HIGHLIGHTS

- ✓ Operated from 230 V, 50 Hz A.C. mains, or by hand cranking
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Anti-parallax mirror scale with knife-edge pointer (not provided on triple-range and above)
- ✓ Guard ring around the line terminal, plus a guard terminal
- ✓ ON-OFF switch, jewel lamp for mains ON and safety fuse
- ✓ Accurate and quick response: 3 seconds
- ✓ Collapsible generator handle
- ✓ Seasoned wooden cabinet with sunmica finish, detachable lid, carrying handle and lock
- ✓ Highly sensitive cross-coil indicator
- ✓ Externally accessible infinity adjuster
- ✓ Terminals accept standard cable terminations, including 4 mm plug and spade
- ✓ Conforms to IS 2992:1987

## 02 TYPICAL APPLICATIONS

- Field insulation testing where mains supply is uncertain or unavailable
- Sustained PI and dielectric absorption testing (motor-driven mode)
- Remote substation and railway traction equipment commissioning
- HV transformer and cable insulation assessment



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/777 HM Technical Catalogue.

## 03 PRINCIPLE AND CONSTRUCTION

### PRINCIPLE

A built-in motor operated from 230 V, 50 Hz A.C. mains, a static rectifier and a cross-coil ohm meter. The instrument can also be operated by hand cranking.

### CONSTRUCTION

Housed in a seasoned wooden cabinet with sunmica finish, complete with detachable lid, carrying handle and lock. Terminals are designed to accommodate standard cable terminations, including 4 mm plug and spade. The weight is 12 kg including top cover (CIE/777: 9 kg).

## 04 SALIENT FEATURES

**Mains motor drive**

A built-in motor operated from 230 V, 50 Hz A.C. mains, with a static rectifier and a cross-coil ohm meter.

**Guard ring and guard terminal**

A guard ring around the line terminal and a guard terminal eliminate errors due to surface leakage. An externally accessible infinity adjuster is also provided.

**Controls and protection**

A highly insulated panel carries an ON-OFF switch, a jewel lamp indicating mains ON and a safety fuse protecting the motor, with a 3-pin socket for the supply cord.

**Hand-crank operation**

The instrument operates on A.C. mains as well as by hand cranking. A centrifugal clutch (governor) holds the output voltage constant, and the generator handle is collapsible.

## 05 PERFORMANCE AND TEST SPECIFICATION

| | |
|---|---|
| Standard | IS 2992:1987 |
| Supply | 230 V, 50 Hz A.C. mains, or hand cranking |
| Indicator | Highly sensitive cross-coil indicator; anti-parallax mirror scale with knife-edge pointer (not provided on triple-range and above instruments) |
| Scale length | 100 mm (approx.) |
| Accuracy | As per IS 2992:1987 |
| Response time | 3 seconds |
| High voltage test | Withstands minimum 2 kV or (rated voltage + 1) kV A.C. (r.m.s.) for 1 minute between the electrical circuit and the containing case |
| Insulation resistance of the instrument | Better than 50 MΩ at 500 V D.C. between the electrical circuit and the containing case |



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/777 HM Technical Catalogue.

## 06 CONSTRUCTION AND PHYSICAL SPECIFICATION

| | |
|---|---|
| Hand operation | Collapsible generator handle; centrifugal clutch (governor) for constant output voltage |
| Guard and infinity adjuster | Guard ring around the line terminal and a guard terminal; externally accessible infinity adjuster |
| Panel | Highly insulated panel with ON-OFF switch, jewel lamp indicating mains ON and safety fuse protecting the motor; 3-pin socket for the supply cord |
| Supply cord | 2 m, with 3-pin plug |
| Terminals | Designed to accommodate standard cable terminations, including 4 mm plug and spade |
| Body | Seasoned wooden cabinet with sunmica finish, detachable lid, carrying handle and lock |
| Weight (including top cover) | 12 kg (approx.) |
| Dimensions (including top cover) | 14 × 10 × 10 inch (approx.) |

## 07 RANGE OPTIONS BY RATED VOLTAGE

Each instrument is built for **one rated voltage** and **one insulation range**. Choose a rated voltage, then **one** range from its row. Ranges are not combined in one instrument unless one of the dual voltage or dual insulation range options below is selected.

| RATED VOLTAGE | INSULATION RESISTANCE RANGE (MΩ): ANY ONE OF |
|---|---|
| 500 V | 0–1000 |
| 1000 V | 0–2000 |
| 2500 V | 0–5000 OR 0–10,000 OR 0–20,000 |
| 5000 V | 0–5000 OR 0–10,000 OR 0–20,000 OR 50–50,000 OR 100–100,000 |
| 10,000 V | 0–10,000 OR 0–20,000 OR 50–50,000 OR 100–100,000 |

*Any other range can be manufactured against specific requirements.*

## 08 DUAL VOLTAGE AND DUAL INSULATION RANGE OPTIONS

### Dual voltage and dual insulation range

Two rated voltages, each with its own insulation range. The selected range must be within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

### Single voltage but dual insulation range

One rated voltage with two insulation ranges, within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

### Dual voltage, single insulation range

Two rated voltages sharing one insulation range, within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/777 HM Technical Catalogue.

## 09 MULTIRANGE MOTOR-CUM-HAND DRIVEN INSULATION TESTER: CIE/777 HM/MR

| SERIES | RATED VOLTAGE | INNER SCALE (MΩ) | OUTER SCALE (MΩ) |
|---|---|---|---|
| **MR-A** | 5000 V | 0–5000 | 50–100,000 |
| | 2500 V | 0–2500 | 25–50,000 |
| | 1000 V | 0–1000 | 10–20,000 |
| **MR-B** | 5000 V | 0–2000 | 10–10,000 |
| | 2500 V | 0–1000 | 5–5000 |
| | 1000 V | 0–500 | 2–2000 |
| **MR-C** | 5000 V | 100–100,000 | Single scale |
| | 2500 V | 50–50,000 | Single scale |
| **MR-D** | 5000 V | 0–5000 | 50–100,000 |
| | 2500 V | 0–2500 | 25–50,000 |
| **MR-E** | 5000 V | 50–50,000 | Single scale |
| | 2500 V | 25–25,000 | Single scale |
| | 1000 V | 10–10,000 | Single scale |

*Multirange instruments are provided with a range selector switch, so one instrument covers every voltage and range listed in its series. Any other range can be manufactured against specific requirements.*

## 10 STANDARD AND OPTIONAL ACCESSORIES

**Supplied as standard**

- ✓ Mains cord, 2 m
- ✓ 1 pair of H.V. testing leads

**Optional, at extra cost**

- Carrying case
- 15-minute timer facility
- Testing leads, 1 m or 2 m
- Additional / special-grade H.V. testing leads, 2 m

## 11 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/777 HM, 500 V / 0–1000 MΩ**. Any range not listed can be manufactured against specific requirements.



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/777 HM Technical Catalogue.